T4 - Spectral Decomposition via Projections and Lagrange Interpolation

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Definition (Projection)

Let V be a vector space over $\mathbb{k}$. A linear map $\pi : V \to V$ is called a **projection** if

$$\pi^2 = \pi.$$

Equivalently,

$$V = \ker(\pi) \oplus \operatorname{im}(\pi).$$

Proposition (Spectral Decomposition for Diagonalizable Maps)

Let $f : V \to V$ be diagonalizable with distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Then there exist projections $\pi_1, \dots, \pi_k$ such that

- Orthogonality:

$$\pi_i \pi_j = \begin{cases} 0 & i \neq j \\ \pi_i & i = j \end{cases}$$

- Resolution of identity:

$$\pi_1 + \cdots + \pi_k = \mathrm{id}_V$$

- Direct sum decomposition:

$$V = W_1 \oplus \cdots \oplus W_k, \quad W_i = \operatorname{im}(\pi_i)$$

- Spectral decomposition:

$$f = \lambda_1 \pi_1 + \cdots + \lambda_k \pi_k \tag{*}$$

Proof

Since f is diagonalizable, we have

$$V = \bigoplus_{i=1}^{k} E_{\lambda_i},$$

where $E_{\lambda_i} = \ker(f - \lambda_i I)$.

Define π_i to be the projection onto E_{λ_i} along $\bigoplus_{j \neq i} E_{\lambda_j}$. Then:

- $\pi_i^2 = \pi_i$ (projection),
- $\pi_i \pi_j = 0$ for $i \neq j$,
- $\sum \pi_i = \mathrm{id}_V$.

For $v = \sum v_i$ with $v_i \in E_{\lambda_i}$,

$$f(v) = \sum \lambda_i v_i = \sum \lambda_i \pi_i(v),$$

which yields $(*)$.

Proposition (Polynomial Functional Calculus)

For any $p \in \Bbbk[x]$,

$$p(f) = p(\lambda_1)\pi_1 + \cdots + p(\lambda_k)\pi_k.$$

Proof

First verify for $p(x) = x^n$:

From $(*)$,

$$f^n = \sum \lambda_i^n \pi_i,$$

using $\pi_i \pi_j = 0$ for $i \neq j$ and $\pi_i^2 = \pi_i$.

Extend by linearity to all polynomials.

Remark

This reduces operator computations to scalar computations on eigenvalues.

Proposition (Lagrange Construction of Projections)

Define, for each λ_i,

$$P_{\lambda_i}(x) = \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j}.$$

Then

$$P_{\lambda_i}(\lambda_j) = \begin{cases} 1 & j = i \\ 0 & j \neq i \end{cases}$$

and

$$\pi_i = P_{\lambda_i}(f).$$

Proof

By polynomial functional calculus,

$$P_{\lambda_i}(f) = \sum_j P_{\lambda_i}(\lambda_j)\pi_j = \pi_i.$$

Example (Explicit Computation)

Let

$$A = \begin{bmatrix} 5 & -6 & -6 \\ 1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}.$$

Then

$$\chi_A = (x-2)^2(x-1), \quad m_A = (x-2)(x-1),$$

so A is diagonalizable.

There exist projections E, E' such that

$$E + E' = I, \quad EE' = 0 = E'E, \quad A = 2E + 1E'.$$

Construct Lagrange polynomials:

$$P_2(x) = \frac{x-1}{2-1} = x - 1, \quad P_1(x) = \frac{x-2}{1-2} = 2 - x.$$

Thus

$$E = P_2(A) = A - I, \quad E' = P_1(A) = 2I - A.$$

Check:

$$EE' = (A - I)(2I - A) = -(A - I)(A - 2I) = 0$$

since $m_A(A) = 0$.

Proposition (Rank-One Projection)

Let $A \in \mathbb{k}^{n\times n}$ with $\operatorname{rank}(A) = 1$. Then

$$A = uv^\top$$

for some $u, v \in \mathbb{k}^n$.

Moreover, A is a projection ($A^2 = A$) if and only if

$$v^\top u = 1.$$

Proof

Compute:

$$A^2 = (uv^\top)(uv^\top) = (v^\top u), uv^\top.$$

Thus $A^2 = A$ iff

$$(v^\top u - 1)uv^\top = 0.$$

Since $\operatorname{rank}(A) = 1$, $uv^\top \neq 0$, hence $v^\top u = 1$.

Proposition (Difference of Projections)

Let $f, g : V \to V$ be projections. Then

$$f - g \,\text{is a projection} \iff \ker(f) \subseteq \ker(g) \,\text{and}\dim(g) \subseteq \operatorname{im}(f).$$

Proof

($\Rightarrow$)

Assume $(f - g)^2 = f - g$. Expanding:

$$(f-g)^2 = f^2 - fg - gf + g^2 = f - fg - gf + g.$$

Equating with $f-g$ yields constraints implying

$$fg = g, \quad gf = g,$$

which give the inclusions.

($\Leftarrow$)

Assume $\ker(f) \subseteq \ker(g)$ and $\mathrm{im}(g) \subseteq \mathrm{im}(f)$.

Write $v = x + y$ with $x \in \ker(f)$, $y \in \mathrm{im}(f)$.

Then

$$(f-g)(v) = y - g(y).$$

Applying again:

$$(f-g)^2(v) = (f-g)(y - g(y)) = y - g(y),$$

using the assumptions. Hence $(f-g)^2 = f - g$.

Remark (Kernel and Image)

Under the above conditions,

$$\ker(f-g) = \ker(f) \oplus \mathrm{im}(g),$$

$$\mathrm{im}(f-g) = \mathrm{im}(f) \cap \ker(g).$$

Conclusion / Summary

- Diagonalizable operators admit a **canonical decomposition** into projections.
- These projections are **polynomials in the operator**, constructed via Lagrange interpolation.
- Polynomial functional calculus reduces operator analysis to eigenvalues.
- Rank-one projections have a precise algebraic form $uv^\top$ with normalization $v^\top u = 1$.
- Differences of projections encode inclusion relations between kernels and images.

This framework is fundamental: it bridges **spectral theory**, **algebraic structure**, and **computational techniques** in linear algebra.